

QUIZ

Why do we maximize **log likelihood** rather than **likelihood** directly while computing a maximum likelihood estimator?

(INFORMAL) COURSE EVALUATION

How can we improve the course for the remainder of the semester?

Don't write your name on the paper.

1- Lectures in terms of:

a- Density b- Slides c- Anything else

2- Homeworks in terms of:

a- Difficulty b- Length c- Fair evaluation d- Anything else

3- SI sessions in terms of:

a- Usefulness b- Content c- Anything else

4- Office hours / Piazza in terms of:

a- Office hour times b- Piazza usefulness c- Anything else

5- Any other suggestions for improving the course.

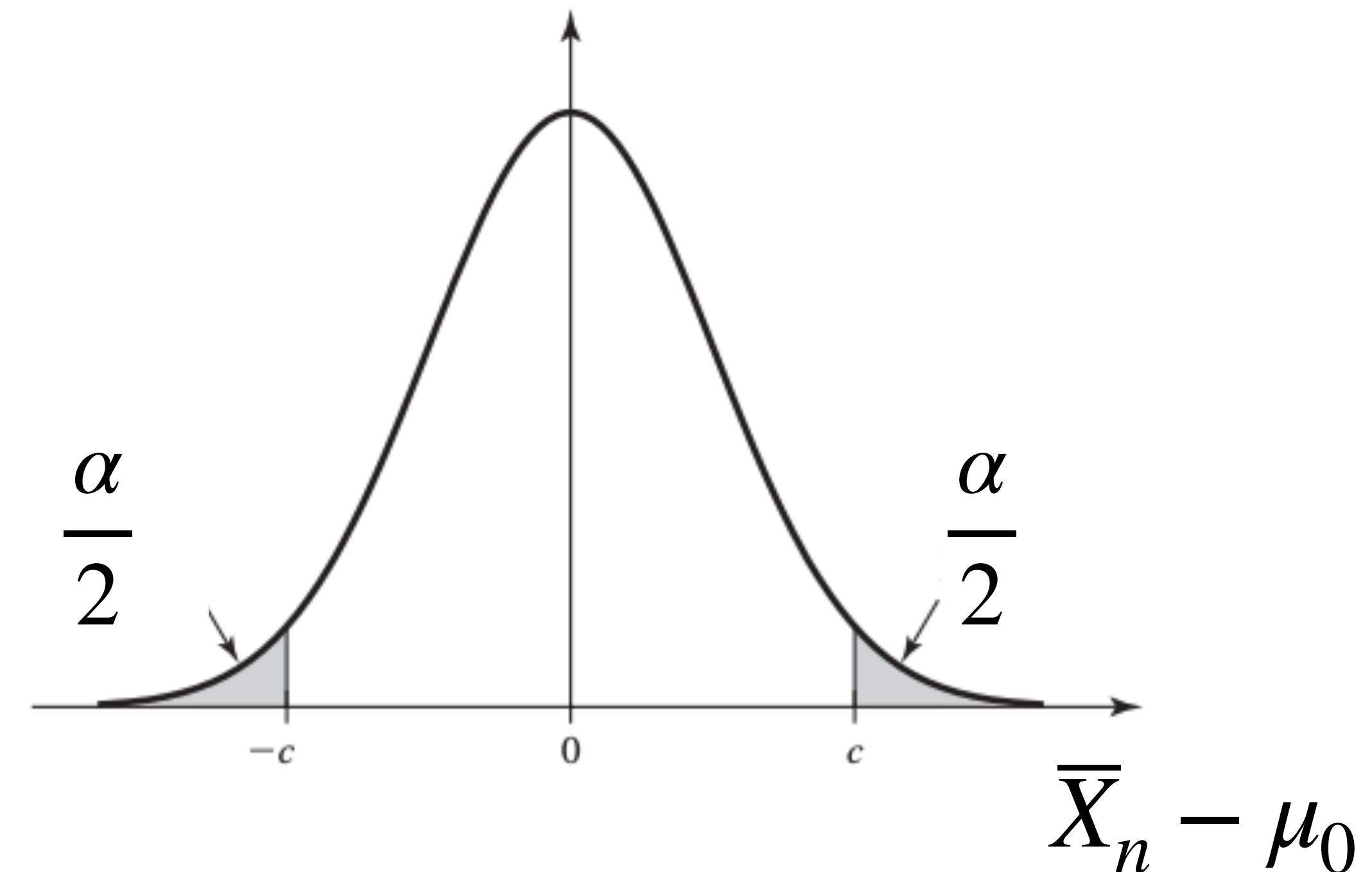
HYPOTHESIS TESTING

Hypothesis regarding mean of a normal distribution with known σ :

Test $H_0 : \mu = \mu_0$ $H_A : \mu \neq \mu_0$

$T = |\bar{X}_n - \mu_0|$, Significance level α : $P_{H_0}(T \geq c) \leq \alpha$

Reject H_0 if $\bar{X}_n - \mu_0$ lies in shaded region.



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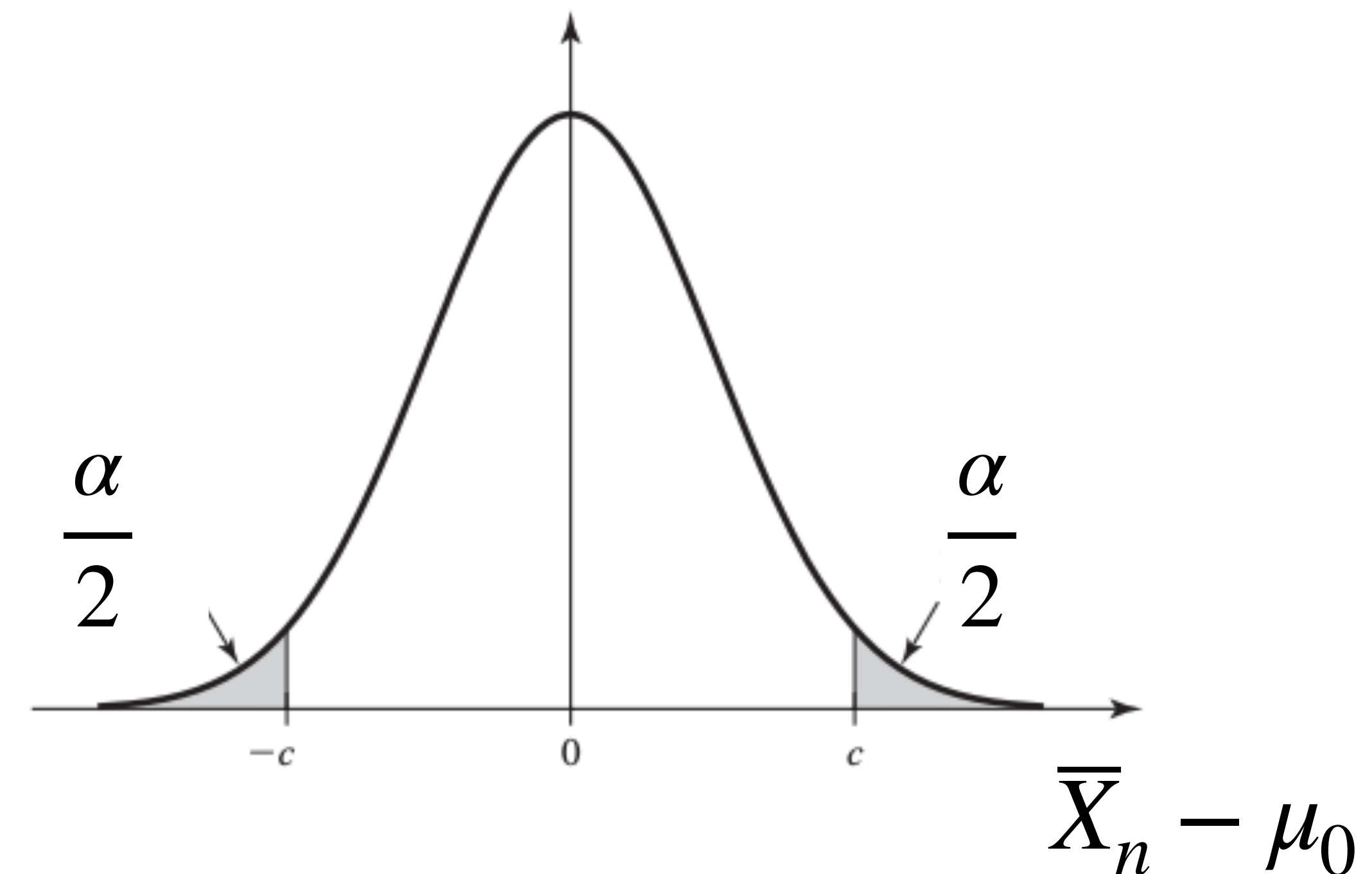
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$$c = \Phi^{-1}(1 - \alpha/2) \frac{\sigma}{\sqrt{n}}$$



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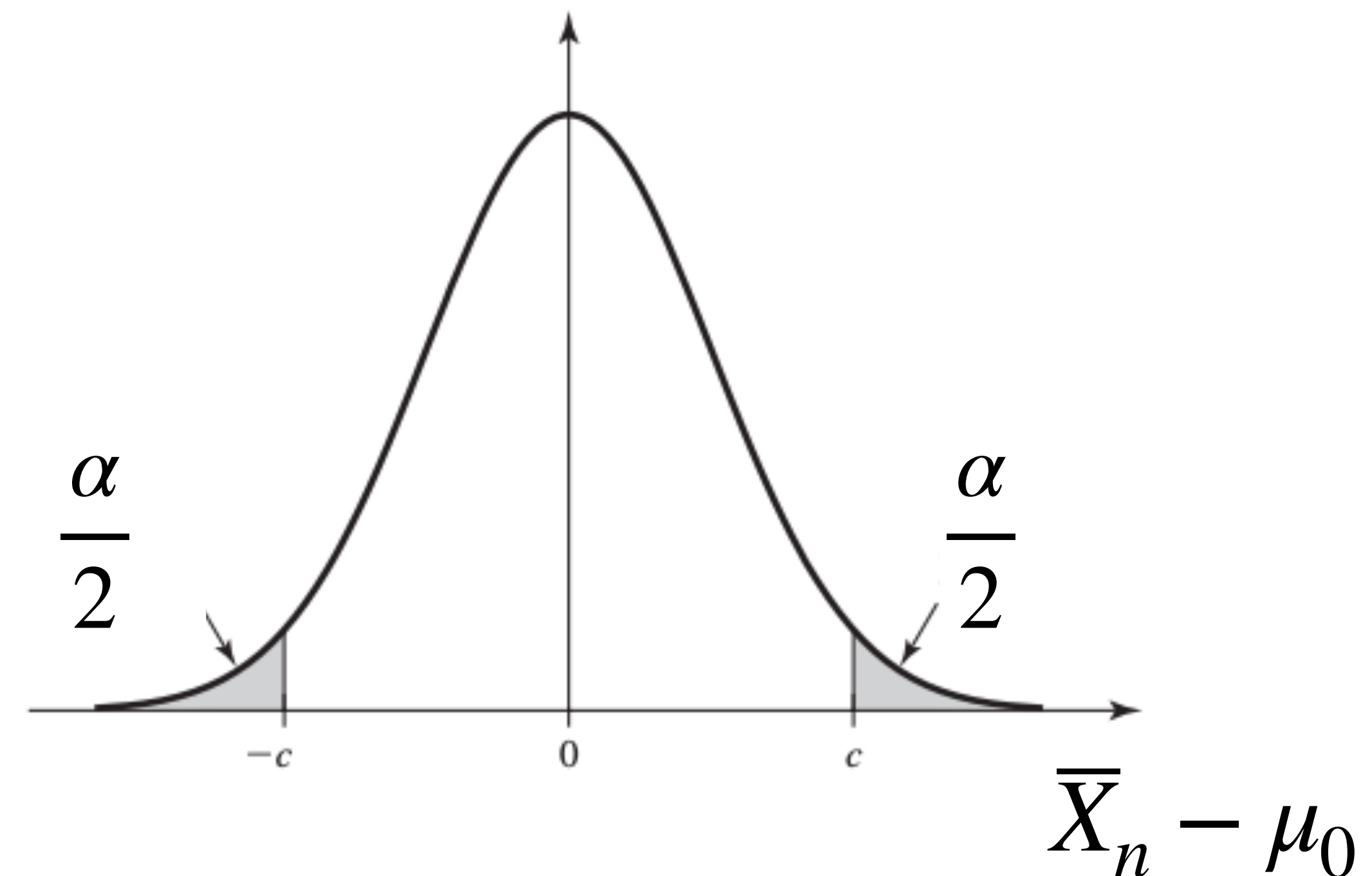
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Calculate T for sample $\bar{X}_n$

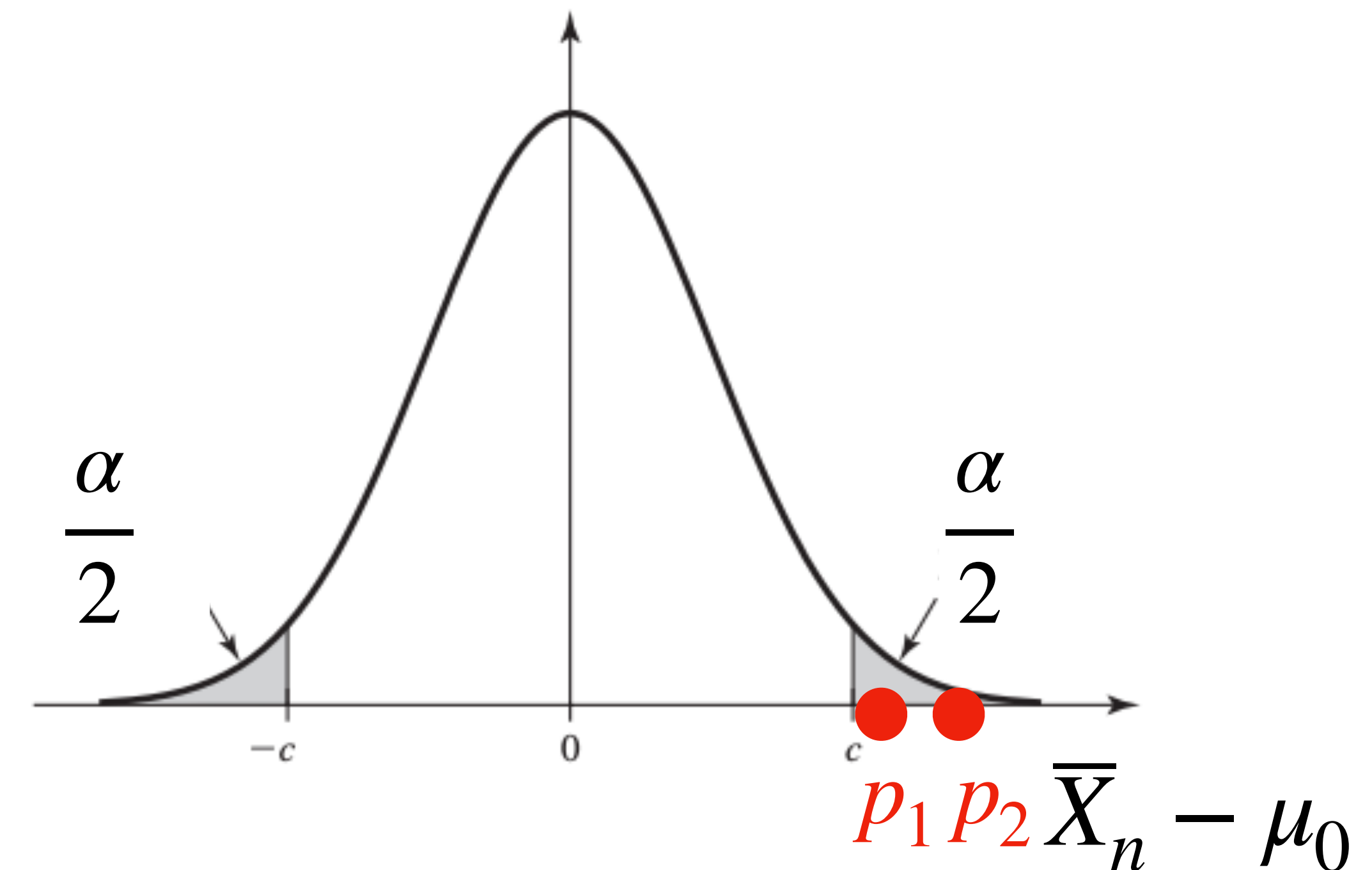
If $T \geq c$ reject H_0 at significance α
else do not reject H_0 .



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Note: No difference between p_1, p_2 :

‘ H_0 rejected at significance level α ’ in both cases. Not so good!



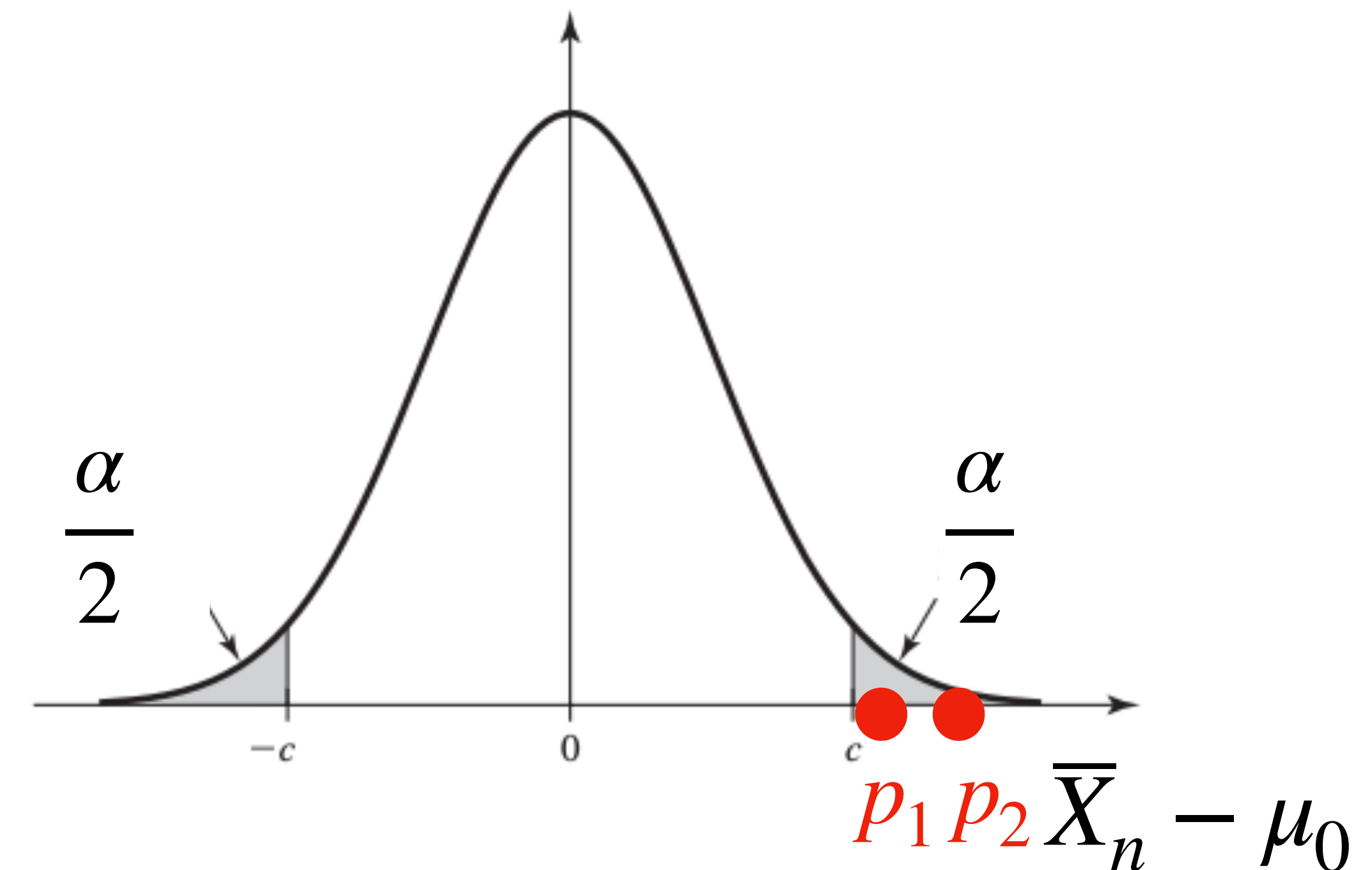
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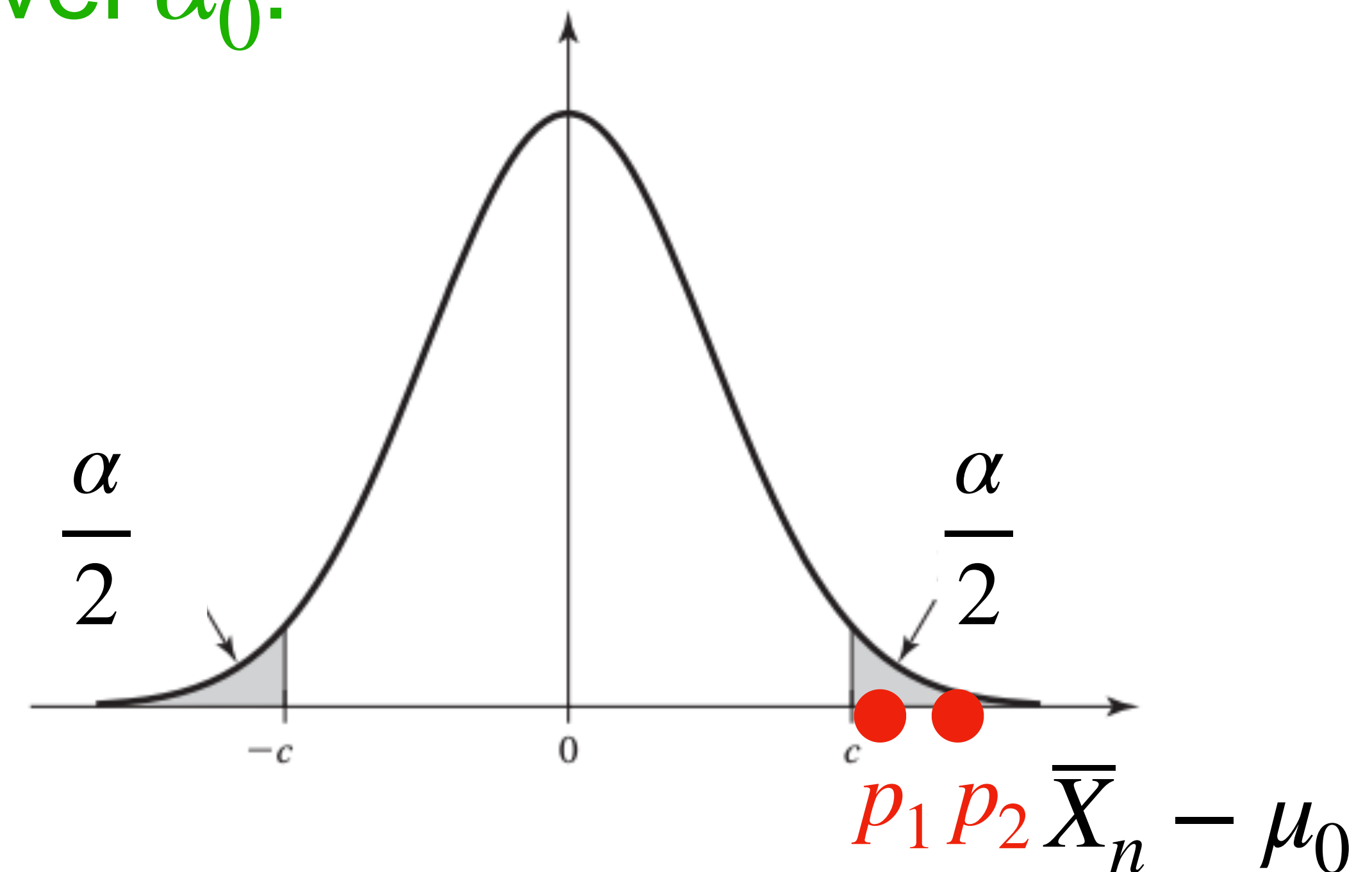
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How? For this example:

$$Z = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma}, \text{ say } Z = 2.78. \quad \text{Smallest } \alpha_0 \text{ s.t. } 2.78 \geq \Phi^{-1}(1 - \alpha_0/2)$$

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```
from scipy.stats import norm
2*(1-norm.cdf(2.78))
```

```
0.005435889845402553
```

p value = 0.0054. Since p value ≤ 0.05
we reject H_0 at significance 0.05.

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For this example, for instance:

Reject H_0 if:

$$\text{Test } H_0 : \mu = \mu_0 \quad H_A : \mu \neq \mu_0 \quad \left| \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} \right| \geq \Phi^{-1}(1 - \alpha/2)$$

$$\text{Test } H_0 : \mu \geq \mu_0 \quad H_A : \mu < \mu_0 \quad \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} \leq \Phi^{-1}(\alpha)$$

$$\text{Test } H_0 : \mu \leq \mu_0 \quad H_A : \mu > \mu_0 \quad \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} \geq \Phi^{-1}(1 - \alpha)$$

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For this example, for instance:

Test $H_0 : \mu = \mu_0$ $H_A : \mu \neq \mu_0$ p value $= 2(1 - \Phi(\frac{\sqrt{n} |\bar{X}_n - \mu_0|}{\sigma}))$

Test $H_0 : \mu \geq \mu_0$ $H_A : \mu < \mu_0$ p value $= \Phi(\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma})$

Test $H_0 : \mu \leq \mu_0$ $H_A : \mu > \mu_0$ p value $= 1 - \Phi(\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma})$

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Ex: # of medical in-patient days in nursing homes.

Sample of $n = 18$ normal random variables, unknown μ and σ^2 .

$\bar{X}_n = 182.17$, $\hat{\sigma}_1 = 72.22$. Test $H_0: \mu \geq 200$, $H_A: \mu < 200$

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Already seen $\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\hat{\sigma}_1}$ has *t-distribution* with $n - 1$ dof.

Observed value: $\frac{\sqrt{18}(182.7 - 200)}{72.22} = -1.018$

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Alternative-1 to present results:

```
from scipy.stats import t
t.ppf(0.05, 17)
```

-1.7396067260750676

Since observed value = $-1.018 > -1.73$ we don't reject H_0 .